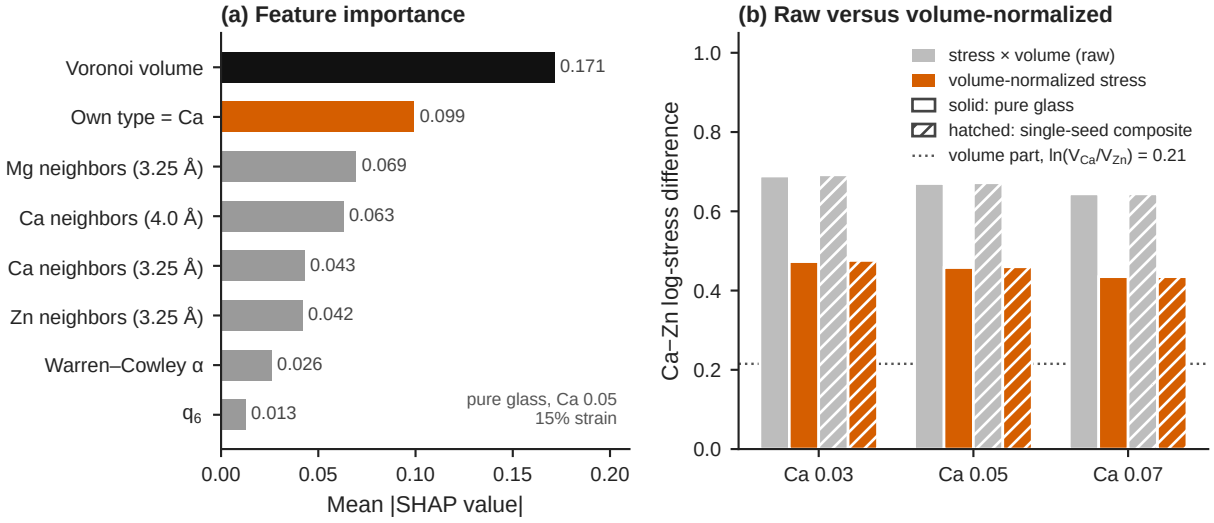
(a) Feature importance
Voronoi volume
Own type = Ca
Mg neighbors (3.25 Å)
Ca neighbors (4.0 Å)
Ca neighbors (3.25 Å)
Zn neighbors (3.25 Å)
Warren–Cowley α
q_6
0.171
0.099
0.069
0.063
0.043
0.042
0.026
0.013
pure glass, Ca 0.05
15% strain
0.00
0.05
0.10
0.15
0.20
Mean |SHAP value|
(b) Raw versus volume-normalized
stress × volume (raw)
volume-normalized stress
solid: pure glass
hatched: single-seed composite
volume part, $\ln(V_{Ca}/V_{Zn}) = 0.21$
Ca−Zn log-stress difference
1.0
0.8
0.6
0.4
0.2
0.0
Ca 0.03
Ca 0.05
Ca 0.07